

MBA: 8109 RESEARCH METHODS



Week 5: Formulating the Research Design

Prof. JO Onyango, PhD

Links: [ORCID](#) | [LinkedIn](#) | [Institutional Profile](#) | [Google Scholar](#)

Email: jonyango@strathmore.edu

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Learning Objectives

STRATEGY

Select & Justify

Select and justify an appropriate research strategy — e.g. survey, case study, action research — for a given research question and objectives.

CHOICES

Decide the Method Mix

Decide between mono-, mixed- and multi-method designs, and defend that choice against the alternatives.

TIME HORIZON

Set the Horizon

Determine the appropriate time horizon for the study — cross-sectional (snapshot) vs longitudinal (diary).

CREDIBILITY

Assess Rigour

Assess the credibility of a proposed design — its reliability, validity, and generalisability — and anticipate threats.

By the end of this session, the candidate should be able to move confidently from a defined research question to a defensible research design.

Where Design Sits in the Research Onion

Philosophy	Positivism · Realism · Interpretivism · Pragmatism
Approach	Deductive · Inductive · Abductive
Strategy	Experiment · Survey · Case Study · Action Research · Grounded Theory · Ethnography · Archival
Choices	Mono method · Multi-method · Mixed methods
Time Horizon	Cross-sectional · Longitudinal
Techniques & Procedures	Data collection and analysis — the centre of the onion

- Design vs tactics: design is the overall plan (this session); tactics are the fine detail of data collection and analysis (Weeks 6–10).

Key Research Design Types — The Vocabulary You Need

BY DATA TYPE

Quantitative Design

Numerical data; tests relationships/hypotheses statistically. Deductive in spirit — typically surveys or experiments.

Qualitative Design

Non-numerical data (words, images, observations); explores meaning, process and context. Inductive in spirit — case study, ethnography, grounded theory.

Mixed Methods Design

Deliberately combines quantitative and qualitative data/analysis in one study to answer different facets of the same question (detail on Slide 10).

BY PURPOSE

Exploratory Design

Used when the problem is not yet clearly defined; seeks insight via literature, expert interviews, focus groups.

Descriptive Design

Portrays an accurate profile of persons, events or situations — the who/what/where/when; requires a clear prior picture of the phenomenon.

Explanatory (Causal) Design

Establishes and tests causal relationships between variables — the why; often builds on exploratory + descriptive work.

BY TIME DIMENSION

Cross-Sectional Design

A 'snapshot' of a phenomenon at one point in time — the most common design given course time constraints.

Longitudinal Design

A 'diary' over an extended period — tracks change and development; independent of strategy or method choice.

Case Study Design

In-depth study of one or a few cases in real context; run as a snapshot or over time, so it sits on either time dimension.

- Not mutually exclusive — a single study sits on all three axes at once, e.g. a cross-sectional, quantitative, descriptive survey design.

Seven Research Strategies, One Principle

Strategy	Core Idea	Typically Used For
Experiment	Study causal links between variables via control/manipulation	Explanatory
Survey	Collect standardised data from a sizeable population/sample	Descriptive, Exploratory
Case Study	In-depth, empirical inquiry into a phenomenon in its real-life context	Explanatory, Exploratory
Action Research	Research in action, with practitioners, to resolve an organisational issue	Explanatory
Grounded Theory	Theory built inductively/deductively from continual reference to data	Exploratory, Explanatory
Ethnography	Describe and explain a social world from the perspective of its members	Exploratory
Archival Research	Uses administrative records and documents as the principal data source	Descriptive, Explanatory

- None is inherently superior; each can serve exploratory, descriptive or explanatory purposes — the fit to your RQ and objectives is what matters.

Experiment, Survey & Quasi-Experiment

STRATEGY

Experiment

Roots in the natural sciences; the 'gold standard' against which other strategies are judged for rigour.

Best for: testing whether change in an independent variable produces change in a dependent variable.

Shortfalls: rare in pure form in management research; artificial settings limit ecological validity; ethical and access constraints on manipulating real variables.

STRATEGY

Survey

Collects standardised data from a sizeable population, usually via questionnaire; associated with the deductive approach and cross-sectional design.

Best for: descriptive and explanatory questions across many cases; relationships between variables.

Shortfalls: depth is traded for breadth; response and non-response bias; standardisation can miss context and nuance.

STRATEGY

Quasi-Experiment

Experimental logic without random allocation; the groups compared are formed by existing conditions rather than by the researcher.

Best for: testing cause and effect in real organisations — before-and-after measures in one unit against a similar comparison unit.

Shortfalls: groups may differ at the outset, so rival explanations are harder to rule out and causal claims are weaker than in a true experiment.

Case Study & Action Research

STRATEGY

Case Study

In-depth inquiry into a phenomenon within its real-life context, especially where context and phenomenon are not clearly separable; single, multiple, holistic or embedded designs.

Best for: exploring or challenging existing theory; generating new research questions.

Shortfalls: often (unfairly) seen as 'unscientific'; limited statistical generalisability; access and time-intensive.

STRATEGY

Action Research

Research in action — an iterative spiral of diagnosing, planning, taking action, and evaluating, in collaborative partnership with practitioners.

Best for: resolving live organisational issues where findings must inform practice beyond the immediate project.

Shortfalls: researcher is embedded in the setting (dual insider role); findings are context-bound; requires sustained organisational access and buy-in.

Grounded Theory, Ethnography & Archival Research

STRATEGY

Grounded Theory

Theory built through continual, iterative reference to data — an inductive/deductive hybrid, not a mechanical procedure.

Shortfall: demands considerable experience and time; ‘messy’ by nature; easily reduced to word counts or theory-testing if misapplied.

STRATEGY

Ethnography

Inductive, naturalistic immersion in a social world to describe and explain it as members do.

Shortfall: very time-consuming; requires sustained access and trust; researcher role and bias hard to separate from the setting.

STRATEGY

Archival Research

Uses administrative records and documents (historical or recent) as the principal data source.

Shortfall: bound by what records exist and their original purpose; quality and completeness of records vary; limited insight into intent behind the data.

How to Select a Research Strategy

Criteria to weigh

- Fit to the research question(s) and objectives — the primary test.
- The extent of existing knowledge in the area (well-mapped territory vs frontier).
- The amount of time and other resources available to you.
- Your own philosophical underpinnings (positivist, interpretivist, pragmatist — Week 4).

Key Principle

Strategies are not mutually exclusive.

- It is quite possible to use a survey strategy as part of a case study.
- What matters is not the label attached to a strategy, but whether it lets you answer your research question(s) and meet your objectives.
- Every design decision — site, unit, respondents — must be defensible: be ready to say why this organisation, why this department, why this group.
- Justification must trace back to the research question/objectives and stay consistent with your research philosophy.

Research Choices: Mono, Multi- and Mixed Methods



Why this matters for your Concept Paper

- This is your Chapter 3 ‘Research Choices’ decision — name it explicitly and justify it against the criteria on the previous slide, not just against convenience.
- Multi-method ≠ mixed methods: multi-method stays within one paradigm (all-quant or all-qual); mixed methods deliberately combines both.
- Increasingly advocated in business and management research — but the choice must still serve the research question, not be adopted for its own sake.

Using Mixed Methods — Why, When, and With Caution

Why choose mixed methods

- Different methods serve different purposes in one study — e.g. interviews to explore issues, then a questionnaire to test/describe them at scale.
- Better opportunities to answer the research question and to evaluate how far findings can be trusted (triangulation).
- Cancels out the ‘method effect’: since every technique shapes the results it produces, combining methods builds confidence in conclusions.
- Can surface findings and uses the researcher had not anticipated (Bryman, 2006).

Bryman’s caution — use it deliberately, not by default

- In a review of 200+ mixed-methods articles, Bryman (2006) found the stated rationale for combining methods often did not match how the methods were actually used.
- Only 10 of the articles clearly showed quant and qual each designed to answer a distinct, specified research question.
- Shortfall: mixing methods multiplies the chance of unanticipated, hard-to-reconcile outcomes if the rationale isn’t made explicit up front.
- Practice discipline: state which method answers which sub-question before you collect data — not after.

Cross-Sectional vs Longitudinal

THE 'SNAPSHOT'

Cross-Sectional

Studies a phenomenon at one particular point in time.

Most course-based research is cross-sectional due to time constraints; often paired with a survey strategy, but qualitative case studies conducted over a short window are cross-sectional too.

THE 'DIARY'

Longitudinal

Studies change and development over an extended period; the researcher can observe how variables move over time.

Stronger for studying causal processes and change — e.g. a longitudinal case study of one company's manufacturing process over a year.

Key point

- Time horizon is independent of your strategy and method choice — you can run a longitudinal case study, a cross-sectional survey, or a longitudinal survey (panel).
- Choice should follow the research question: is change over time part of what you need to explain, or is a defensible snapshot sufficient?

Reliability

Reliability = the extent to which your data collection or analysis procedures yield consistent findings.

Test it by asking: (1) Will the measures yield the same results on other occasions? (2) Will similar observations be reached by other observers? (3) Is there transparency in how sense was made from the raw data?

THREAT 1

Subject / Participant Error

Results shift with when/how data is collected — e.g. a workplace-enthusiasm survey answered differently on a Friday afternoon than a Monday morning.

THREAT 2

Subject / Participant Bias

Respondents say what they think is expected of them — common under authoritarian management or job-insecurity conditions.

THREAT 3

Observer Error

Different researchers ask questions differently — mitigated by a more structured interview schedule.

THREAT 4

Observer Bias

Different researchers interpret the same responses differently — a threat to consistency in coding and analysis.

Validity & Generalisability

Validity = whether the findings are really about what they appear to be about — is the relationship you observe genuinely causal?

History	An external event around the time of data collection distorts findings (e.g. a product recall just before a customer-opinion study).
Testing	Awareness of being studied changes participants' behaviour or answers.
Instrumentation	A change in procedure between data-collection rounds (e.g. new instructions to staff) biases later results.
Mortality	Participants drop out of the study, skewing what remains.
Maturation	Other events unfold over the study period and confound the effect you are studying.
Ambiguous causal direction	Unclear which variable is causing which — e.g. does a negative attitude cause poor performance ratings, or vice versa?

Generalisability (external validity)

- The extent to which findings apply beyond the study sample/context — typically limited for case studies and small, purposively-sampled qualitative work; a key trade-off to name and defend, not hide.

Shortfalls at a Glance — A Quick-Reference Table

Design	Key Strength	Key Shortfall
Experiment	Precision; strong causal claims	Artificial; rarely feasible in pure form in management research
Survey	Breadth; comparability across cases	Depth traded for scale; response/non-response bias
Case Study	Rich, contextual, real-life insight	Limited statistical generalisability; resource-intensive
Action Research	Directly useful to practice; iterative learning	Researcher embeddedness; findings context-bound
Grounded Theory	Theory genuinely built from data	Time- and skill-intensive; easily misapplied
Ethnography	Deep, member's-eye understanding	Very time-consuming; hard to separate bias from immersion
Archival Research	Access to real, unobtrusive records	Bound by what records exist and their original purpose
Mixed Methods	Triangulation; cancels 'method effect'	Needs an explicit rationale, or risks unanticipated, unreconciled findings

Design Clinic — Map Your Question to a Strategy

The task

- In pairs/triads, candidates bring their working research question and objectives.
- Map the question to one (or a justified combination of) research strategy.
- State: philosophy → approach → strategy → method choice → time horizon.
- Name the strongest shortfall of the chosen design, and how you will mitigate it.
- Be ready to defend the choice against at least one alternative strategy.

Design Justification Checklist

- Does the strategy fit my research question and objectives?
- What does existing knowledge in this area already tell me?
- Do I have the time and resources this design requires?
- Is this design consistent with my stated research philosophy?
- Have I justified my method choice (mono / multi / mixed) against the alternatives — not just convenience?
- Have I set a time horizon deliberately, not by default?
- Have I named the main threat to reliability/validity and how I will address it?

Next session — Week 6: Sampling Techniques and Sample Size Decisions